

Kontsevich Deformation Quantization in Lean

Problem statement and a formalization path

Track 4 working note

August 12, 2026

Abstract

This note states the deformation quantization problem solved by Kontsevich's formality theorem, fixes notation for every structure involved, and lays out a layered path for formalizing the result in Lean 4 / Mathlib. The note is intentionally minimal. It records what must be defined, what is already available in Mathlib, what is missing, and which subproblems can be attacked first. References are cited by paper, section, and result number throughout.

1 Background and goal

Let M be a smooth manifold. Write T_M for its tangent bundle:

$$T_M := \bigsqcup_{p \in M} T_p M,$$

where $T_p M$ is the tangent vector space at p . A smooth vector field is a section of T_M , i.e. an element of $\Gamma(T_M)$. A bivector field is a section of $\Lambda^2 T_M$, i.e. an element of $\Gamma(\Lambda^2 T_M)$.

A *Poisson manifold* is a pair (M, π) where M is a smooth manifold and $\pi \in \Gamma(\Lambda^2 T_M)$ is a bivector field whose Schouten–Nijenhuis self-bracket vanishes:

$$[\pi, \pi]_{\text{SN}} = 0.$$

This condition is equivalent to the Jacobi identity for the bracket

$$\{f, g\} := \pi(df, dg), \quad f, g \in C^\infty(M).$$

Thus $\{\cdot, \cdot\}$ is a Poisson bracket on $C^\infty(M)$. In local coordinates, if

$$\pi = \frac{1}{2} \sum_{i,j} \pi^{ij}(x) \partial_i \wedge \partial_j,$$

then $[\pi, \pi]_{\text{SN}} = 0$ is equivalent to

$$\sum_{\ell} \left(\pi^{i\ell} \partial_\ell \pi^{jk} + \pi^{j\ell} \partial_\ell \pi^{ki} + \pi^{k\ell} \partial_\ell \pi^{ij} \right) = 0 \quad \text{for all } i, j, k.$$

The problem posed by physics is to associate, to each (M, π) , an associative noncommutative product $\star$ on $C^\infty(M)[[\hbar]]$ whose semiclassical limit is π . The precise notion is a star product (Bayen–Flato–Fronsdal–Lichnerowicz–Sternheimer [1]).

Definition 1.1. A *star product* on (M, π) is an $\mathbb{R}[[\hbar]]$ -bilinear map

$$\star: C^\infty(M)[[\hbar]] \otimes_{\mathbb{R}[[\hbar]]} C^\infty(M)[[\hbar]] \longrightarrow C^\infty(M)[[\hbar]], \quad f \star g = fg + \sum_{k \geq 1} \hbar^k B_k(f, g),$$

such that:

- (i) each B_k is a bidifferential operator on M ;
- (ii) $\star$ is associative;
- (iii) $B_1(f, g) - B_1(g, f) = \{f, g\}$.

Two star products $\star, \star'$ are *gauge equivalent* if there exists $T = \text{id} + \sum_{k \geq 1} \hbar^k T_k$ with T_k differential operators and $T(f \star g) = T f \star' T g$.

Example 1.2 (Standard Poisson structure). Let $\mathbb{R}^{2m}$ have coordinates $(q_1, \dots, q_m, p_1, \dots, p_m)$. The bivector field

$$\pi_0 := \sum_{i=1}^m \frac{\partial}{\partial q_i} \wedge \frac{\partial}{\partial p_i}$$

is Poisson. Its coefficients are constant, hence $[\pi_0, \pi_0]_{\text{SN}} = 0$. The induced bracket is

$$\{f, g\} = \sum_{i=1}^m \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right).$$

Example 1.3 (The Moyal product). Fix an antisymmetric constant matrix $\pi^{ij} = -\pi^{ji}$ on $\mathbb{R}^d$ and set $\pi = \frac{1}{2} \sum_{i,j} \pi^{ij} \partial_i \wedge \partial_j$. The Moyal product is

$$f \star g = m \left(\exp \left(\frac{\hbar}{2} \sum_{i,j} \pi^{ij} \partial_i \otimes \partial_j \right) (f \otimes g) \right),$$

where $m(f \otimes g) = fg$. Its first terms are

$$f \star g = fg + \frac{\hbar}{2} \sum_{i,j} \pi^{ij} (\partial_i f)(\partial_j g) + O(\hbar^2),$$

so

$$f \star g - g \star f = \hbar \{f, g\} + O(\hbar^2).$$

The goal of Kontsevich's formality theorem is to show that such a $\star$ exists for every Poisson structure, and to classify them up to gauge equivalence by Poisson structures up to formal diffeomorphism.

2 The two graded Lie algebras

Kontsevich's framework expresses the problem as a comparison of two L_∞ -algebras built from M .

2.1 Polyvector fields

Definition 2.1. For a smooth manifold M , set

$$T_{\text{poly}}^n(M) := \Gamma(\Lambda^{n+1} T_M), \quad n \geq -1.$$

Equipped with the Schouten–Nijenhuis bracket $[\cdot, \cdot]_{\text{SN}}$ and zero differential, $T_{\text{poly}}(M)$ is a graded Lie algebra concentrated in degrees ≥ -1 .

A Maurer–Cartan element of $T_{\text{poly}}(M)[[\hbar]]$ in degree 1 is an element $\pi = \hbar \pi_1 + \hbar^2 \pi_2 + \dots$ with $\pi_k \in \Gamma(\Lambda^2 T_M)$ and $[\pi, \pi]_{\text{SN}} = 0$.

Example 2.2 (Polyvector fields on $\mathbb{R}^2$). For $M = \mathbb{R}^2$ with coordinates (x, y) ,

$$T_{\text{poly}}^{-1}(M) = C^\infty(M), \quad T_{\text{poly}}^0(M) = \Gamma(T_M), \quad T_{\text{poly}}^1(M) = \Gamma(\Lambda^2 T_M).$$

A vector field is, for example,

$$X = x\partial_x + y\partial_y.$$

A bivector field is, for example,

$$\pi = f(x, y)\partial_x \wedge \partial_y.$$

Every bivector field on a surface is Poisson, because $[\pi, \pi]_{\text{SN}}$ is a trivector field and $\Lambda^3 T_M = 0$.

Example 2.3 (Polyvector fields on the Riemann sphere). Let $M = S^2 \cong \mathbb{CP}^1$ viewed as a real 2-manifold. Cover M by two stereographic charts:

$$U_N = M \setminus \{S\}, \quad z = x + iy \in \mathbb{C}; \quad U_S = M \setminus \{N\}, \quad w = u + iv \in \mathbb{C};$$

with transition $w = 1/z$ on $U_N \cap U_S$.

The tangent bundle T_M is a real rank-2 vector bundle, locally trivialised by (∂_x, ∂_y) on U_N and by (∂_u, ∂_v) on U_S . As a bundle over S^2 it is nontrivial: every smooth global vector field on S^2 has a zero (Hairy Ball Theorem). The polyvector-field grading is

$$T_{\text{poly}}^{-1}(M) = C^\infty(M), \quad T_{\text{poly}}^0(M) = \Gamma(T_M), \quad T_{\text{poly}}^1(M) = \Gamma(\Lambda^2 T_M),$$

and $T_{\text{poly}}^n(M) = 0$ for $n \geq 2$ because $\Lambda^k T_p M = 0$ for $k > \dim_{\mathbb{R}} M = 2$.

The bundle $\Lambda^2 T_M$ has real rank $\binom{2}{2} = 1$; a global section is a bivector field. On the two charts,

$$\pi|_{U_N} = f(x, y)\partial_x \wedge \partial_y, \quad \pi|_{U_S} = g(u, v)\partial_u \wedge \partial_v,$$

with $f(x, y) = \det J(x, y) \cdot g(u(x, y), v(x, y))$ on the overlap, where $J = \partial(u, v)/\partial(x, y)$ is the Jacobian of the change of coordinates.

A canonical choice of $\pi \in \Gamma(\Lambda^2 T_M)$ comes from the Fubini–Study Kähler form

$$\omega_{\text{FS}} = \frac{i}{2} \frac{dz \wedge d\bar{z}}{(1 + |z|^2)^2} = \frac{dx \wedge dy}{(1 + x^2 + y^2)^2},$$

inverted to the bivector

$$\pi_{\text{FS}}|_{U_N} = (1 + x^2 + y^2)^2 \partial_x \wedge \partial_y, \quad \pi_{\text{FS}}|_{U_S} = (1 + u^2 + v^2)^2 \partial_u \wedge \partial_v.$$

Both local expressions come from inverting the same intrinsic ω_{FS} , so they glue on $U_N \cap U_S$.

Poisson-ness is automatic on S^2 : for any $\pi \in \Gamma(\Lambda^2 T_M)$, $[\pi, \pi]_{\text{SN}} \in \Gamma(\Lambda^3 T_M) = 0$ by the same dimension argument. So (S^2, π_{FS}) is a Poisson manifold, with bracket

$$\{f, g\}(z) = (1 + |z|^2)^2 (\partial_x f \cdot \partial_y g - \partial_y f \cdot \partial_x g)$$

on U_N . This is the Poisson bracket of the Fubini–Study symplectic structure and the input data for a Kontsevich star product on S^2 .

The holomorphic tangent bundle $T_{\mathbb{CP}^1}^{1,0} \cong \mathcal{O}(2)$ has complex rank 1, so $\Lambda^2 T_{\mathbb{CP}^1}^{1,0} = 0$. There is no non-trivial holomorphic Poisson structure on $\mathbb{CP}^1$. The Kontsevich framework as stated in Section 1 uses the real tangent bundle for this reason.

2.2 Polydifferential operators

Definition 2.4. Let $A := C^\infty(M)$. Define

$$D_{\text{poly}}^n(M) := \{ D : A^{\otimes(n+1)} \rightarrow A \mid D \text{ is polydifferential} \}, \quad n \geq -1.$$

The Hochschild differential $d_H = [\mu, \cdot]_G$, where μ is the pointwise product, and the Gerstenhaber bracket $[\cdot, \cdot]_G$ make $D_{\text{poly}}(M)$ into a differential graded Lie algebra (Gerstenhaber [5, §6]).

A Maurer–Cartan element of $\hbar D_{\text{poly}}(M)[[\hbar]]$ in degree 1 is $\pi = \hbar B_1 + \hbar^2 B_2 + \dots$ with each B_k bidifferential and

$$d_H \pi + \frac{1}{2} [\pi, \pi]_G = 0.$$

This equation is equivalent to associativity of the deformed product $B := \mu + \pi = \mu + \hbar B_1 + \hbar^2 B_2 + \dots$. So MC elements of D_{poly} are exactly star products.

Example 2.5 (A polydifferential operator). Let $A = C^\infty(\mathbb{R})$. The map

$$D(f, g) = f'g''$$

is a bidifferential operator, hence an element of $D_{\text{poly}}^1(\mathbb{R})$. The ordinary product $\mu(f, g) = fg$ is also an element of $D_{\text{poly}}^1(M)$. The Gerstenhaber square measures the associator:

$$\frac{1}{2} [\mu, \mu]_G(f, g, h) = (fg)h - f(gh),$$

so $[\mu, \mu]_G = 0$ is equivalent to associativity.

2.3 The HKR map

Theorem 2.6 (Hochschild–Kostant–Rosenberg). *The map $U_{\text{HKR}} : T_{\text{poly}}(M) \rightarrow D_{\text{poly}}(M)$ defined by*

$$U_{\text{HKR}}(\xi_1 \wedge \dots \wedge \xi_n)(f_1, \dots, f_n) := \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \xi_{\sigma(1)}(f_1) \dots \xi_{\sigma(n)}(f_n)$$

is a quasi-isomorphism of cochain complexes.

The map U_{HKR} is *not* a morphism of dg Lie algebras. The failure is the source of the problem. Kontsevich’s theorem upgrades U_{HKR} to a structure that does respect the brackets, in a relaxed sense.

Example 2.7 (HKR on a decomposable bivector). For vector fields X, Y ,

$$U_{\text{HKR}}(X \wedge Y)(f, g) = \frac{1}{2} (X(f)Y(g) - Y(f)X(g)).$$

Thus a bivector field becomes an antisymmetric bidifferential operator.

3 L_∞ -algebras and the formality theorem

Definition 3.1. An L_∞ -algebra on a graded $\mathbb{K}$ -module L is a family of graded antisymmetric maps

$$Q_n : L^{\wedge n} \rightarrow L, \quad |Q_n| = 2 - n, \quad n \geq 1,$$

satisfying the generalized Jacobi identities, for every $n \geq 1$,

$$\sum_{i+j=n+1} \sum_{\sigma \in \text{Sh}(i, n-i)} \text{sgn}(\sigma) \varepsilon(\sigma) Q_j(Q_i(x_{\sigma(1)}, \dots, x_{\sigma(i)}), x_{\sigma(i+1)}, \dots, x_{\sigma(n)}) = 0,$$

where $\varepsilon(\sigma)$ is the Koszul sign and $\text{Sh}(i, n-i)$ is the set of $(i, n-i)$ -shuffles.

A differential graded Lie algebra $(L, d, [\cdot, \cdot])$ is the special case $Q_1 = d$, $Q_2 = [\cdot, \cdot]$, $Q_{n \geq 3} = 0$.

Example 3.2 (A dg Lie algebra as an L_∞ -algebra). If $(L, d, [\cdot, \cdot])$ is a differential graded Lie algebra, then it becomes an L_∞ -algebra by setting

$$Q_1 = d, \quad Q_2 = [\cdot, \cdot], \quad Q_n = 0 \quad (n \geq 3).$$

The $n = 1$ identity says $d^2 = 0$. The $n = 2$ identity says that d is compatible with the bracket. The $n = 3$ identity says the graded Jacobi identity.

Definition 3.3. An L_∞ -morphism $U: (L, Q) \rightarrow (L', Q')$ is a family of graded antisymmetric maps

$$U_n: L^{\wedge n} \rightarrow L', \quad |U_n| = 1 - n, \quad n \geq 1,$$

satisfying the compatibility identities of [8, §3.4]. The first component U_1 is a chain map $(L, Q_1) \rightarrow (L', Q'_1)$. A morphism U is a *quasi-isomorphism* if U_1 is.

Definition 3.4. A *Maurer–Cartan element* of an L_∞ -algebra (L, Q) over $\hbar\mathbb{K}[[\hbar]]$ is $\alpha \in L^1 \otimes \hbar\mathbb{K}[[\hbar]]$ such that

$$\sum_{n \geq 1} \frac{1}{n!} Q_n(\alpha, \dots, \alpha) = 0.$$

Example 3.5 (Maurer–Cartan in a dg Lie algebra). For a dg Lie algebra, the Maurer–Cartan equation is

$$d\alpha + \frac{1}{2}[\alpha, \alpha] = 0.$$

For $T_{\text{poly}}(M)$ the differential is zero, so the equation becomes $[\pi, \pi]_{\text{SN}} = 0$. Thus Poisson bivectors are exactly Maurer–Cartan elements of $T_{\text{poly}}(M)$. For $D_{\text{poly}}(M)$, the same equation is associativity of the deformed product.

An L_∞ -morphism U transports MC elements: if α is MC in L , then $\sum_{n \geq 1} \frac{1}{n!} U_n(\alpha, \dots, \alpha)$ is MC in L' .

Theorem 3.6 (Kontsevich formality [8, Thm. 6.4]). *For every smooth manifold M there exists an L_∞ -quasi-isomorphism*

$$U: T_{\text{poly}}(M) \longrightarrow D_{\text{poly}}(M)$$

whose first Taylor component U_1 equals U_{HKR} of Theorem 2.6.

Corollary 3.7 (Quantization). *For every Poisson structure π on M , the formal series*

$$f \star g := fg + \sum_{n \geq 1} \frac{\hbar^n}{n!} U_n(\pi, \dots, \pi)(f, g)$$

defines a star product on (M, π) , and the assignment $\pi \mapsto \star$ induces a bijection between formal Poisson structures modulo formal diffeomorphism and star products modulo gauge equivalence.

Example 3.8 (How formality produces the star product). For a Poisson bivector π , Kontsevich's maps U_n produce

$$f \star g = fg + \sum_{n \geq 1} \frac{\hbar^n}{n!} U_n(\pi, \dots, \pi)(f, g).$$

The L_∞ identities say that the condition $[\pi, \pi]_{\text{SN}} = 0$ is transported to the associativity of $\star$.

4 The graphical formula on $\mathbb{R}^d$

For $M = \mathbb{R}^d$ Kontsevich constructs U by an explicit sum over graphs. This is the formula that has to be implemented.

Definition 4.1 (Kontsevich admissible graph [8, §6.1]). For $n, m \geq 0$, an *admissible graph* Γ of type (n, m) is a finite directed graph with:

- vertex set $V(\Gamma) = V_1 \sqcup V_2$, where $V_1 = \{1, \dots, n\}$ are *aerial* vertices and $V_2 = \{\bar{1}, \dots, \bar{m}\}$ are *ground* vertices;
- each aerial vertex has exactly two outgoing edges, labelled $(\cdot, 1)$ and $(\cdot, 2)$;
- ground vertices have no outgoing edges;
- no edge ends at its starting vertex; no two edges share both endpoints.

Write $G_{n,m}$ for the (finite) set of admissible graphs of type (n, m) . Each $\Gamma \in G_{n,m}$ has $2n$ edges.

Remark 4.2. The definition above matches Kontsevich's bivector case: every aerial vertex has outdegree 2, so B_Γ takes n bivector fields as inputs. The general formality theorem allows aerial vertices of arbitrary outdegree $a_k + 1$, with input a polyvector in $\Gamma(\Lambda^{a_k+1} T_M)$; in that case $|E(\Gamma)| = \sum_k (a_k + 1)$. We restrict to outdegree 2 throughout because it suffices to construct the star product on a Poisson manifold.

Example 4.3 (The first admissible graph). Take $n = 1$ and $m = 2$. There is one aerial vertex 1 and two ground vertices $\bar{1}, \bar{2}$. Let the two outgoing edges from 1 end at $\bar{1}$ and $\bar{2}$. This graph is the first-order graph in the Kontsevich formula.

Definition 4.4 (Polydifferential operator B_Γ). For $\Gamma \in G_{n,m}$, bivector fields $\gamma_1, \dots, \gamma_n$ on $\mathbb{R}^d$, and functions $f_1, \dots, f_m \in C^\infty(\mathbb{R}^d)$, set

$$B_\Gamma(\gamma_1, \dots, \gamma_n)(f_1, \dots, f_m) := \sum_{I: E(\Gamma) \rightarrow \{1, \dots, d\}} \prod_{k=1}^n \left(\prod_{e \in \text{in}(k)} \partial_{I(e)} \right) \gamma_k^{I(e_{k,1}) I(e_{k,2})} \cdot \prod_{j=1}^m \left(\prod_{e \in \text{in}(\bar{j})} \partial_{I(e)} \right) f_j,$$

where $\text{in}(v)$ is the multiset of edges ending at v and $(e_{k,1}, e_{k,2})$ are the two outgoing edges at aerial vertex k .

Example 4.5 (The operator of the first graph). For the graph in Example 4.3 and a bivector $\pi = \frac{1}{2} \sum_{i,j} \pi^{ij} \partial_i \wedge \partial_j$ with $\pi^{ij} = -\pi^{ji}$,

$$B_\Gamma(\pi)(f, g) = \sum_{i,j} \pi^{ij} (\partial_i f)(\partial_j g).$$

The two labelings of the outgoing edges $(1 \rightarrow \bar{1}, 1 \rightarrow \bar{2})$ versus $(1 \rightarrow \bar{2}, 1 \rightarrow \bar{1})$ contribute π^{ij} and π^{ji} , and the antisymmetry of π^{ij} folds them into a single term. This is the first-order bidifferential operator attached to the Poisson bracket.

Definition 4.6 (Configuration space and weight). Let $\mathbb{H} := \{z \in \mathbb{C} : \Im z > 0\}$. For $2n + m - 2 \geq 0$, set

$$C_{n,m} := \{(p_1, \dots, p_n; q_1, \dots, q_m) \in \mathbb{H}^n \times \mathbb{R}^m \mid \text{all distinct, } q_1 < \dots < q_m\} / G_{\text{aff}},$$

where $G_{\text{aff}} = \{z \mapsto az + b : a > 0, b \in \mathbb{R}\}$. Let $\bar{C}_{n,m}$ denote its Fulton–MacPherson–Axelrod–Singer compactification [8, §5], a manifold with corners of dimension $2n + m - 2$.

For an edge $e = (v_1, v_2)$ of an admissible graph and a configuration in $C_{n,m}$, let $\phi_e \in \mathbb{R}/2\pi\mathbb{Z}$ be the hyperbolic angle from the geodesic ray $v_1 \rightarrow \infty$ in the Poincaré model of $\mathbb{H}$ to the geodesic from v_1 to v_2 .

For $\Gamma \in G_{n,m}$ with $|E(\Gamma)| = 2n + m - 2$ define

$$w_\Gamma := \frac{1}{(2\pi)^{2n+m-2}} \int_{\bar{C}_{n,m}^+} \bigwedge_{e \in E(\Gamma)} d\phi_e,$$

where $\bar{C}_{n,m}^+$ is the connected component in which the q_j are ordered $q_1 < \dots < q_m$ on $\mathbb{R}$ and the wedge is taken in a fixed total order on $E(\Gamma)$. For all other Γ set $w_\Gamma := 0$. Since our aerial vertices have outdegree 2, we have $|E(\Gamma)| = 2n$, and the condition $|E(\Gamma)| = 2n + m - 2$ forces $m = 2$: nonzero weights occur only for $\Gamma \in G_{n,2}$.

Example 4.7 (The configuration space $C_{1,2}$). For $n = 1, m = 2$, a configuration consists of one point $p_1 \in \mathbb{H}$ and two ordered real points $q_1 < q_2$. Quotienting by $z \mapsto az + b$ with $a > 0$ lets one normalize

$$q_1 = 0, \quad q_2 = 1.$$

The remaining point p_1 varies in the upper half-plane. The graph weight integrates the angle form over the compactified version of this space.

Theorem 4.8 (Kontsevich’s explicit formality on $\mathbb{R}^d$ [8, §6.4]). *Restricted to bivectors, the maps*

$$U_n(\gamma_1, \dots, \gamma_n) = \sum_{\Gamma \in G_{n,2}} w_\Gamma B_\Gamma(\gamma_1, \dots, \gamma_n) \quad (\gamma_k \in \Gamma(\Lambda^2 T_{\mathbb{R}^d}))$$

extend U_{HKR} and, together with their analogues on graphs with aerial vertices of higher outdegree (Remark 4.2), assemble into an L_∞ -quasi-isomorphism $T_{\text{poly}}(\mathbb{R}^d) \rightarrow D_{\text{poly}}(\mathbb{R}^d)$. The L_∞ identities follow from Stokes’ theorem: for each graph Γ appearing in a given identity, the closed form $\bigwedge_e d\phi_e$ on the compactified configuration space $\bar{C}_{n,m+1}^+$ has an integral that decomposes, via Stokes, into contributions from the codimension-1 boundary strata. These strata correspond bijectively to compositions of two admissible graphs, and the resulting sum is precisely the L_∞ relation.

Example 4.9 (Constant Poisson tensors). If the coefficients π^{ij} are constant, then every graph with an edge landing at an aerial vertex gives zero, because it differentiates a coefficient of π . Only graphs whose edges land on ground vertices survive. In this case Kontsevich’s formula reduces to the Moyal product of Example 1.3.

Example 4.10 (Boundary strata and Stokes’ theorem). When points in a configuration collide, the compactification $\bar{C}_{n,m}$ records the limiting direction of the collision. These codimension-1 boundary faces correspond to compositions of graphs. Stokes’ theorem says that the integral of an exact form over the compactified space is zero. The resulting sum of boundary contributions is precisely the L_∞ identity for the maps U_n .

5 Formalization path in Lean / Mathlib

This section is the working plan for Track 4. The aim is to make Theorem 3.6 a Lean statement and then attack Theorem 4.8 layer by layer.

5.1 Available infrastructure in Mathlib

Audited against Mathlib at commit 360da6fa66 (2026-06-18, toolchain leanprover/lean4: v4.32.0-rc1). The following are in Mathlib at a usable level.

Linear and graded algebra.

- Tensor and exterior algebras of a module, in `Mathlib.LinearAlgebra.TensorAlgebra.Basic` (with grading in `TensorAlgebra.Grading`) and `Mathlib.LinearAlgebra.ExteriorAlgebra.Basic` (grading, basis, and construction from alternating maps in the sibling files).
- Exterior powers $\Lambda^k V$ and alternating multilinear maps as first-class objects, in `Mathlib.LinearAlgebra.ExteriorPower` and `Mathlib.LinearAlgebra.Alternating`.
- Graded monoids, graded rings, graded algebras, and graded modules, in `Mathlib.Algebra.GradedMonoid`, `Mathlib.Algebra.GradedMulAction`, `Mathlib.Algebra.Module.GradedModule`, `Mathlib.Algebra.DirectSum.Algebra`, and `Mathlib.Algebra.DirectSum.Ring`.
- Lie rings, Lie algebras, and Lie modules in `Mathlib.Algebra.Lie.Basic`; graded Lie algebras in `Mathlib.Algebra.Lie.Graded`.
- Derivations of a commutative ring / algebra in `Mathlib.RingTheory.Derivation.Basic` (with the Lie-bracket instance in `Mathlib.RingTheory.Derivation.Lie`).

Formal power series.

- Univariate formal power series (used for the coefficient ring $\mathbb{R}[[\hbar]]$) in `Mathlib.RingTheory.PowerSeries.Basic`, with substitution, order, inverses, and truncation in the sibling files.
- Multivariate power series in `Mathlib.RingTheory.MvPowerSeries.Basic` with substitution and evaluation in the sibling files.

Homological algebra.

- Chain and cochain complexes over an abelian category (`Mathlib.Algebra.Homology.HomologicalComplex`, `Mathlib.Algebra.Homology.CochainComplexPlus`), together with quasi-isomorphisms (`Mathlib.Algebra.Homology.QuasiIso`) and homology functors. This is the dg framework we will wrap dg Lie algebras around.

Smooth manifolds, tangent bundles, vector fields.

- Smooth manifolds via `Mathlib.Geometry.Manifold.IsManifold.Basic`, built on the *ModelWithCorners* abstraction. So manifolds with corners are supported at the level of models. Interior and boundary of such a manifold, and the fact that they partition it, live in `Mathlib.Geometry.Manifold.IsManifold.InteriorBoundary`.
- Tangent bundle as a C^n vector bundle (`Mathlib.Geometry.Manifold.VectorBundle.Tangent`) and smooth sections of a vector bundle (`Mathlib.Geometry.Manifold.VectorBundle.ContMDiffSection`).
- Point derivations of $C^\infty(M)$ and the differential of a smooth map, in `Mathlib.Geometry.Manifold.DerivationBundle`.
- Lie bracket of vector fields on a manifold, in `Mathlib.Geometry.Manifold.VectorField.LieBracket`.

Differential forms and calculus.

- Exterior derivative of a differential form on a normed space in `Mathlib.Analysis.Calculus.DifferentialForm.Basic` (Kudryashov–Lindauer, 2025), with the Cartan-style pairing against vector fields in `DifferentialForm.VectorField`. Only defined on normed spaces so far: the file explicitly lists “bundled C^r -smooth n -forms on manifolds” as a TODO.

5.2 Missing infrastructure

The audit above shows the following are absent from Mathlib. Each gap becomes a workstream in Section 6.

Homotopical algebra.

- L_∞ -algebras and L_∞ -morphisms with Koszul signs. No `LInftyAlgebra`, no `MaurerCartan`, nothing in Mathlib on either.
- Maurer–Cartan elements of a dg Lie or L_∞ -algebra, and gauge equivalence over $\hbar\mathbb{K}[[\hbar]]$.
- Operads and algebras over operads (needed only for the Tamarkin alternative in Section 5.7).

Hochschild theory of associative algebras.

- Hochschild cochain complex of an associative algebra A (multilinear maps $A^{\otimes n} \rightarrow A$ with the Hochschild differential). A repository-wide search returns exactly one mention of “Hochschild”, a comment on the Hochschild–Serre spectral sequence in group cohomology; there is no Hochschild homology or cohomology of associative algebras in Mathlib.
- Gerstenhaber bracket on that complex; the associated dg Lie structure; the fact that MC elements are deformations of the product.

Poisson / polyvector-field side.

- Polyvector fields $\Gamma(\Lambda^{n+1}T_M)$ as a graded $C^\infty(M)$ -module. The pieces exist (exterior power of a module; tangent bundle; smooth sections) but not the graded object $T_{\text{poly}}(M)$.
- Schouten–Nijenhuis bracket on $T_{\text{poly}}(M)$, its Jacobi identity, and the vanishing of $[\cdot, \cdot]_{\text{SN}}$ on functions.
- Poisson manifolds, Poisson brackets, the equivalence between Poisson bivectors and Lie brackets on $C^\infty(M)$. Grep for “Poisson” in `Mathlib/Geometry` returns nothing; the existing hits under `Analysis` are the Poisson kernel and Poisson summation, unrelated.

Polydifferential operators on $C^\infty(M)$.

- Polydifferential operators as a concrete subtype of $C^\infty(M)^{\otimes \bullet} \rightarrow C^\infty(M)$. On $\mathbb{R}^d$ with polynomial coefficients this is a routine algebraic subtype; on a general manifold it requires jet bundles.
- Star products on (M, π) and gauge equivalence between them.

Forms, integration, corners.

- Differential forms on manifolds (Mathlib has them only on normed spaces, per the file `DifferentialForm/Basic.lean` TODO).
- Integration of differential forms on an oriented manifold, including the case of a manifold with corners.
- Stokes’ theorem for manifolds with corners. The abstract model `ModelWithCorners` is available and boundary points are defined, but there is no integration theory that uses them.

Configuration spaces.

- Kontsevich admissible graphs $G_{n,m}$ as a finite decidable type.

- Fulton–MacPherson–Axelrod–Singer compactifications $\bar{C}_{n,m}$, their manifold-with-corners structure, and the identification of codimension-1 boundary strata with compositions of smaller configuration spaces.
- The Kontsevich angle 1-form $d\phi_e$ on $\bar{C}_{n,m}^+$ and the weight integrals w_Γ .

5.3 Task decomposition for the initial team

The six workstreams of Section 5.2 split into atomic tasks. Each task fixes a Lean module path, a mathematical statement, its dependencies among other tasks, and a target sorry count. The sorry-count target is 0 for algebraic tasks and equals the number of documented axiom blocks for tasks that isolate the analytic input. Effort estimates are S (one to two weeks), M (two to four weeks), L (four to eight weeks) for a Lean-fluent contributor. The task graph is laid out so that Workstream A, and the first task of each other workstream, can be started in parallel on day one and produce a visible artefact within two weeks.

Workstream A. Combinatorial and polynomial scaffolding. Foundational Lean-typeclass work. No non-trivial mathematics. Serves as the entry point for contributors calibrating their Lean.

- A1** *Admissible graphs as a decidable finite type.* S . Module `Kontsevich.Combinatorics.AdmissibleGraph`. Implement the structure of Definition 4.1 with `DecidableEq` and a `Fintype` instance on $G_{n,m}$. Deliverable: `#eval` enumerates $G_{1,2}$ (one graph), $G_{2,2}$ (four graphs after edge labelling), and $G_{3,2}$. Depends on: none.
- A2** *Polynomial ring on $\mathbb{R}^d$ as an ambient algebra.* S . Module `Kontsevich.Polynomial.RdPoly`. Wrap `Mathlib.Algebra.MvPolynomial.Basic` with the smooth-function interpretation: coordinate derivations ∂_i as `MvPolynomial.derivation`, and a lemma that they commute pairwise. Depends on: none.
- A3** *Polynomial polyvector fields on $\mathbb{R}^d$.* S . Module `Kontsevich.Polynomial.Tpoly`. Define $T_{\text{poly}}^n(\mathbb{R}^d)_{\text{poly}}$ as $\mathbb{R}[x_1, \dots, x_d] \otimes \Lambda^{n+1} \mathbb{R}^d$ using `Mathlib.LinearAlgebra.ExteriorPower`. Depends on: A2.
- A4** *Polynomial polydifferential operators on $\mathbb{R}^d$.* M . Module `Kontsevich.Polynomial.Dpoly`. Define $D_{\text{poly}}^n(\mathbb{R}^d)_{\text{poly}}$ as multilinear maps $\mathbb{R}[x]^{\otimes(n+1)} \rightarrow \mathbb{R}[x]$ whose action factors as a finite polynomial combination of ∂^α 's. Depends on: A2.
- A5** *The operator B_Γ in polynomial coefficients.* M . Module `Kontsevich.Polynomial.BGamma`. Implement Definition 4.4 for $\gamma_k \in T_{\text{poly}}^1(\mathbb{R}^d)_{\text{poly}}$ and $f_j \in \mathbb{R}[x]$. Depends on: A1, A3, A4.
- A6** *Kontsevich weights as an axiomatised black box.* S . Module `Kontsevich.Analysis.WeightAxiom`. Declare $w: G_{n,2} \rightarrow \mathbb{R}$ as an `axiom` for each n , with the values for $n = 1, 2$ substituted by the closed-form rationals from [8, §8.3]. Weights for $n \geq 3$ stay abstract. Depends on: A1.

Workstream B. Graded and Lie infrastructure. Bridges `Mathlib`'s existing typeclasses to the `dg` Lie structures the project needs. Runs in parallel with A.

- B1** *dg Lie algebra wrapper on Mathlib.* S to M . Module `Kontsevich.Algebra.DGLie`. Assemble a `DGLA` structure from `Mathlib.Algebra.Homology.HomologicalComplex` over the category of vector spaces and `Mathlib.Algebra.Lie.Basic`. Prove Jacobi and the Leibniz rule for d as instance obligations. Depends on: none.

- B2** *Schouten–Nijenhuis bracket on $T_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$.* *M.* Module `Kontsevich.Polynomial`. `Schouten`. Implement $[\cdot, \cdot]_{\text{SN}}$ via the local-coordinate formula, prove graded antisymmetry and the graded Jacobi identity. Deliverable: theorems `schouten_antisymm` and `schouten_jacobi`. Depends on: A2, A3.
- B3** *Gerstenhaber bracket on $D_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$.* *M.* Module `Kontsevich.Polynomial`. `Gerstenhaber`. Implement $[\cdot, \cdot]_{\text{G}}$ via the sum-over-insertions formula ([5], §6), prove graded antisymmetry and graded Jacobi. Depends on: A2, A4.
- B4** *Hochschild differential and $d_{\text{H}}^2 = 0$.* *S.* Module `Kontsevich.Polynomial.Hochschild`. Set $d_{\text{H}} := [\mu, \cdot]_{\text{G}}$ where μ is the pointwise product. Prove $d_{\text{H}}^2 = 0$ by expanding $[[\mu, \mu]_{\text{G}}, \cdot]_{\text{G}}$. Depends on: B3.
- B5** *Two dg Lie instances.* *S.* Module `Kontsevich.Polynomial.DGLieInstances`. Register $T_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$ (with zero differential) and $D_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$ (with differential d_{H}) as DGLA instances via B1. Depends on: B1, B2, B4.

Workstream C. Hochschild–Kostant–Rosenberg. Bridges the two dg Lie algebras of Workstream B. Contains the first non-trivial theorem the team can point at.

- C1** *U_{HKR} as a graded linear map.* *S.* Module `Kontsevich.Polynomial.HKR`. Define $U_{\text{HKR}}: T_{\text{poly}}(\mathbb{R}^d)_{\text{poly}} \rightarrow D_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$ by Theorem 2.6. Depends on: A3, A4.
- C2** *U_{HKR} is a chain map.* *M.* Same module. Prove $d_{\text{H}} \circ U_{\text{HKR}} = 0$ (the domain has zero differential). Depends on: B4, C1.
- C3** *U_{HKR} is a quasi-isomorphism on $\mathbb{R}^d$ -polynomials.* *M to L.* Module `Kontsevich.Polynomial.HKRQuasiIso`. Compute Hochschild cohomology of $\mathbb{R}[x_1, \dots, x_d]$ using the Koszul resolution and match it to $T_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$. Depends on: C2.

Workstream D. L_{∞} machinery and Maurer–Cartan. The sign-heavy homotopical algebra core. Sign conventions are fixed in D1 and cited by label everywhere else in the project.

- D1** *L_{∞} -algebra typeclass with Koszul signs.* *L.* Module `Kontsevich.LInfty.Algebra`. Implement Definition 3.1. Fix the sign convention of [8, §3.4] in an `assumption` block and prove the $n = 1, 2, 3$ special cases. Depends on: none.
- D2** *L_{∞} -morphism typeclass.* *M.* Module `Kontsevich.LInfty.Morphism`. Implement Definition 3.3 and the quasi-isomorphism predicate on U_1 . Depends on: D1.
- D3** *Embedding of dg Lie algebras into L_{∞} -algebras.* *S.* Module `Kontsevich.LInfty.FromDGLA`. Prove that a DGLA is an L_{∞} -algebra with $Q_n = 0$ for $n \geq 3$. Depends on: B1, D1.
- D4** *Maurer–Cartan elements over $\hbar\mathbb{K}[[\hbar]]$.* *M.* Module `Kontsevich.LInfty.MaurerCartan`. Implement Definition 3.4. Prove that MC elements of a dg Lie algebra satisfy $d\alpha + \frac{1}{2}[\alpha, \alpha] = 0$. Depends on: D1.
- D5** *Gauge equivalence of MC elements.* *M to L.* Module `Kontsevich.LInfty.Gauge`. Implement the exponential action of degree-0 elements and prove that gauge equivalence is an equivalence relation on MC elements. Depends on: D4.
- D6** *MC transport under L_{∞} -morphisms.* *M.* Module `Kontsevich.LInfty.MCTransport`. Prove that $\alpha \mapsto \sum_{n \geq 1} \frac{1}{n!} U_n(\alpha^{\wedge n})$ sends MC to MC. Depends on: D2, D4.

Workstream E. Kontsevich formula on $\mathbb{R}^d$. Assembles A–D into the polynomial-coefficient formality statement.

- E1** *The map U_n in polynomial coefficients.* *M.* Module `Kontsevich.Formula.U`. Define $U_n := \sum_{\Gamma \in G_{n,2}} w_\Gamma B_\Gamma$ as a map $(T_{\text{poly}}^1(\mathbb{R}^d)_{\text{poly}})^{\wedge n} \rightarrow D_{\text{poly}}^1(\mathbb{R}^d)_{\text{poly}}$. Prove U_1 equals U_{HKR} restricted to bivectors. Depends on: A5, A6, C1.
- E2** *The Moyal product from the Kontsevich formula.* *M.* Module `Kontsevich.Formula.Moyal`. Specialise E1 to constant π and prove the resulting star product equals the Moyal product of Example 1.3. Prove associativity directly by graph enumeration for $n \leq 3$ and inductively for general n . Depends on: A5, E1.
- E3** *L_∞ identities modulo boundary axioms.* *L.* Module `Kontsevich.Formula.LInftyIds`. Prove that the maps U_n of E1 satisfy the L_∞ identities of D1 modulo an explicit axiom block encoding the boundary identities of Theorem 4.8. Sorry count: 0; axiom count: one per required boundary vanishing. Depends on: B5, D3, D6, E1.
- E4** *Star-product corollary.* *M.* Module `Kontsevich.Formula.StarProduct`. Deduce Corollary 3.7 for polynomial Poisson tensors on $\mathbb{R}^d$: the series $f \star g = fg + \sum_{n \geq 1} \frac{\hbar^n}{n!} U_n(\pi, \dots, \pi)(f, g)$ is associative. Depends on: D6, E3.

Workstream F. Analytic input (phase two). Discharges the axioms of A6 and E3. This workstream is deferred until Workstreams A–E deliver Milestones 1–8 of Section 6; documented here so that the axiom count of E3 is bounded by a concrete list.

- F1** *Configuration space $C_{n,m}$ as a smooth manifold.* *L.* Module `Kontsevich.Analytic.Config`. Define $C_{n,m}$ and equip it with the smooth structure from Definition 4.6. Depends on: A1.
- F2** *The Kontsevich angle 1-form $d\phi_e$.* *M.* Module `Kontsevich.Analytic.AngleForm`. Depends on: F1.
- F3** *Compactification $\bar{C}_{n,m}$ as a manifold with corners.* *L.* Module `Kontsevich.Analytic.Compactify`. Build the Fulton–MacPherson–Axelrod–Singer compactification. Identify the codimension-1 boundary strata. Depends on: F1.
- F4** *Stokes’ theorem on $\bar{C}_{n,m}^+$ and boundary vanishing.* *L.* Module `Kontsevich.Analytic.Stokes`. Discharge the axiom block of E3. Depends on: F2, F3.
- F5** *Weight computation for low n .* *S to M.* Module `Kontsevich.Analytic.WeightsLow`. Replace the axiomatised values of A6 for $n = 1, 2$ with theorems that evaluate the integral w_Γ . Depends on: F2, F3.

Suggested allocation. A team of six to eight contributors of mixed Lean fluency covers the six workstreams. A workable initial mapping: two contributors on Workstream A (foundational Lean, junior friendly), one on B1 shared with two more on B2–B5 (mid-level Lean plus one algebraist per bracket), one on Workstream C (algebraic depth), one to two on Workstream D (sign-heavy, senior Lean), and one shared coordinator on Workstream E. Workstream F is deferred. The task graph has three cut points that gate visible team progress: A5+A6 (a runnable B_Γ on graphs), B2+B3 (the two brackets), and E2 (the Moyal product as a Kontsevich computation). Each cut point yields a self-contained artefact that can be reviewed independently.

5.4 Layered plan

Layer 0 (Combinatorial scaffolding). Implement `AdmissibleGraph` (`n m : Nat`) from Definition 4.1 as a finite, decidable structure. Build the polynomial-coefficient version of B_Γ on $\mathbb{R}^d$ acting on polynomial functions. Axiomatize $w_\Gamma : \mathbb{Q}_\Gamma \rightarrow \mathbb{R}$ as a real-valued constant per graph, postponing its integral definition. This layer is purely finite and decidable. It is testable: for small n, m one can check the Kontsevich formula returns the Moyal product on the constant Poisson structure.

Layer 1. Algebraic L_∞ . Define `LInftyAlgebra` (`V : Type*`) [`GradedModule V`] with the maps Q_n of Definition 3.1 and the generalized Jacobi identities. Define `LInftyMorphism` per Definition 3.3. Construct the dg-Lie embedding $\text{DGLA} \hookrightarrow L_\infty\text{-Alg}$. Define Maurer–Cartan elements and gauge equivalence over $\hbar\mathbb{K}[[\hbar]]$. Sign conventions follow [8, §4]; they must be declared once and reused.

Layer 2. Concrete dg-Lie algebras. For $V = \mathbb{R}^d$, define $T_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$ as polynomial polyvector fields with the Schouten–Nijenhuis bracket; this avoids analysis. Define $D_{\text{poly}}(\mathbb{R}^d)_{\text{poly}}$ as polydifferential operators with polynomial coefficients, with the Gerstenhaber bracket and Hochschild differential. Prove U_{HKR} is a chain map (Theorem 2.6) in this polynomial setting.

Layer 3. Formality. With the axiomatized weights of Layer 0 and the algebraic infrastructure of Layers 1–2, prove that the maps U_n of Theorem 4.8 define an L_∞ -morphism modulo a single hypothesis: the boundary integrals on $\bar{C}_{n,m+1}^+$ vanish in the combinations required by the L_∞ identities. The hypothesis is the only analytic input; everything else is finite combinatorics with rational coefficients.

Layer 4. Discharging the analytic hypothesis. Two routes are available.

- (a) Build the integration theory: compactified configuration spaces, integration of forms on manifolds with corners, Stokes’ theorem with corners. Then prove the boundary vanishing directly.
- (b) Replace the analytic input by an axiom block citing [8, §6] and document the gap. This is the recommended starting point.

5.5 Subproblems that can be closed without analysis

Two reduced problems are fully algebraic and can be formalized today:

- *Linear Poisson structures on $\mathfrak{g}^*$.* The restriction of U to linear Poisson tensors on the dual of a finite-dimensional Lie algebra $\mathfrak{g}$ equals the symmetrization map $\text{Sym}(\mathfrak{g}) \rightarrow \mathcal{U}(\mathfrak{g})$ composed with the Duflo isomorphism (Kathotia [6, Thm. 1.1], Shoikhet [10]). The PBW theorem and the Duflo factor can be formalized without configuration-space integrals.
- *The Moyal product.* For constant Poisson structures on $\mathbb{R}^d$, all weights w_Γ with Γ having edges among aerial vertices vanish, and the formula reduces to the exponential of $\hbar\pi^{ij}\partial_i \otimes \partial_j/2$. This is the natural first end-to-end verification target.

5.6 Challenges

- *Sign conventions.* Koszul signs in the L_∞ identities are the dominant source of errors in any formalization of homotopical algebra. They must be fixed in Layer 1 and never revisited.

- *Transcendental weights.* The numbers w_Γ are periods of differential forms on manifolds with corners. Some are known closed form (e.g. wheel weights are computed in [10]); most are not. Axiomatization is required unless one ports the integration theory.
- *Polydifferential operators.* A faithful Lean type for polydifferential operators on $C^\infty(M)$ requires a careful treatment of jet bundles or local coordinate charts. Starting with polynomial coefficients on $\mathbb{R}^d$ side-steps this.
- *Gauge equivalence.* The classification statement in Corollary 3.7 requires gauge equivalence of MC elements modulo $\hbar$, which requires the full L_∞ machinery, including the homotopy transfer needed for equivalences.
- *Globalization.* Kontsevich’s $\mathbb{R}^d$ formula globalizes to manifolds by Fedosov-style methods (Cattaneo–Felder–Tomassini [3], Dolgushev [4]). This is a second-stage project. It depends on a connection / formal geometry layer in Mathlib that does not yet exist.

5.7 Alternative approaches

- *Tamarkin’s proof* [11]. Replaces the configuration-space integrals by the formality of the little disks operad E_2 and a choice of Drinfeld associator. This is more algebraic and avoids analysis, but requires operad theory in Mathlib, which is also missing.
- *Cattaneo–Felder Poisson sigma model* [2]. Gives w_Γ as expectation values of a topological field theory on the disk. Not directly formalizable, but a useful guide for interpreting the boundary identities of Theorem 4.8.
- *Manchon–Torossian* [9]. Refines the tangent cohomology of the Kontsevich star product and gives sharper statements at the level of cup products. A target for formalization once the base layer is in place.

6 Concrete first milestones

The proposed sequence of milestones, each independently verifiable:

1. `AdmissibleGraph` type with decidable equality and enumeration; unit tests against [8, §6.1].
2. `LInftyAlgebra` structure with Koszul signs; the embedding $\text{DGLA} \hookrightarrow L_\infty\text{-Alg}$.
3. Schouten–Nijenhuis bracket on polynomial polyvector fields on $\mathbb{R}^d$; Jacobi identity proved.
4. Gerstenhaber bracket on polynomial polydifferential operators on $\mathbb{R}^d$; Hochschild differential proved square-zero.
5. Theorem 2.6 as a Lean theorem in the polynomial setting.
6. Moyal product as the Kontsevich formula evaluated on constant π ; associativity proved by graph enumeration.
7. PBW-symmetrization formula for linear π on $\mathfrak{g}^*$; Duflo factor handled as an axiom on first pass.
8. Statement of Theorem 3.6 with axiomatized weights; the L_∞ identities reduced to the boundary hypothesis.

Milestones 1–6 are fully algebraic and have no analytic axioms. Milestone 7 has one axiom (the Duflo factor) that is independently documentable. Milestone 8 is the cleanest target for a first paper.

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